till10.14-deadline10.20

Chapter 3.8 (P174)

Q2. Suppose that a random variable X can have each of the seven values $-3, -2, -1, 0, 1, 2, 3$ with equal probability. Determine the *p.f.* of $Y = X^2 - X$.

Q3. Suppose that the *p.d.f.* of a random variable X is as follows:

$$f(x) = \begin{cases} \frac{1}{2}x & \text{for } 0 < x < 2, \\ 0 & \text{otherwise.} \end{cases}$$

Also, suppose that $Y = X(2 - X)$. Determine the *c.d.f.* and the *p.d.f.* of Y.

Q14. Let X have the uniform distribution on the interval $[a, b]$, and let $c > 0$. Prove that $cX + d$ has the uniform distribution on the interval $[ca + d, cb + d]$.

Q^{add1}. Suppose that r.v.$X \sim N(\mu, \sigma^2)$, proving that r.v. $Y = \frac{X - \mu}{\sigma}$ obeys the standard normal distribution.

Chapter 3.4 (P129)

Q2. Suppose that in an electric display sign there are three light bulbs in the first row and four light bulbs in the second row. Let X denote the number of bulbs in the first row that will be burned out at a specified time t, and let Y denote the number of bulbs in the second row that will be burned out at the same time t. Suppose that the joint *p.f.* of X and Y is as specified in the following table:

	Y				
X	0	1	2	3	4
0	0.08	0.07	0.06	0.01	0.01
1	0.06	0.10	0.12	0.05	0.02
2	0.05	0.06	0.09	0.04	0.03
3	0.02	0.03	0.03	0.03	0.04

Determine each of the following probabilities:

a. Pr(X =2) b. Pr(Y ⩾2) c. Pr(X <2 and Y ⩽2)

d.Pr(X =y) e. Pr(X > Y)

Q4. Suppose that X and Y have a continuous joint distribution for which the joint *p.d.f.* is defined as follows:

$$f(x, y) = \begin{cases} cy^2 & \text{for } 0 \le x \le 2 \text{ and } 0 \le y \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

Determine (a) the value of the constant c; (b) Pr($X+Y>2$); (c)Pr($Y<1/2$); (d) Pr($X \leqslant 1$); (e) Pr($X=3Y$).

Q8. Suppose that X and Y are random variables such that(X ,Y) must belong to the rectangle in the xy-plane containing all points (x, y) for which 0<x⩽3 and 0<y⩽4. Suppose also that the joint c.d.f. of X and Y at every point(x, y) in this rectangle is specified as follows:

$$F(x, y) = \frac{1}{156} xy(x^2 + y).$$

Determine (a) Pr(1⩽X ⩽2 and 1<Y⩽2); (b) Pr(2⩽X⩽4 and 2<Y⩽4); (c) the c.d.f. of Y;(d) the joint p.d.f. of X and Y; (e) Pr(Y⩽X).

Chapter 3.5 (P140)

Q3. Suppose that X and Y have a continuous joint distribution for which the joint *p.d.f.* is defined as follows:

$$f(x, y) = \begin{cases} \frac{3}{2}y^2 & \text{for } 0 \le x \le 2 \text{ and } 0 \le y \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

a. Determine the marginal *p.d.f.*'s of X and Y.

b. Are X and Y independent?

c. Are the event $\{X<1\}$ and the event $\{Y \geqslant 1/2\}$ independent?

Q9. Suppose that a point (X, Y) is chosen at random from the rectangle S defined as follows: $S = \{(x,y) : 0 \le x \le 2 \text{ and } 1 \le y \le 4\}$.

a. Determine the joint *p.d.f.* of X and Y, the marginal *p.d.f.* of X, and the marginal *p.d.f.* of Y.

b. Are X and Y independent?

Q10. Suppose that a point (X, Y) is chosen at random from the circle S defined as follows: $S = \{(x,y): x^2 + y^2 \le 1\}$.

a. Determine the joint *p.d.f.* of X and Y, the marginal *p.d.f.* of X, and the marginal *p.d.f.* of Y.

b. Are X and Y independent?

Q6. Suppose that in a certain drug the concentration of a particular chemical is a random variable with a continuous distribution for which the *p.d.f.* g is as follows:

$$g(x) = \begin{cases} \frac{3}{8}x^2 & \text{for } 0 \le x \le 2, \\ 0 & \text{otherwise.} \end{cases}$$

Suppose that the concentrations X and Y of the chemical in two separate batches of the drug are independent random variables for each of which the *p.d.f.* is g. Determine(a) the joint *p.d.f.* of X and Y; (b) $\Pr(X = Y)$; (c)$\Pr(X > Y)$;(d) $\Pr(X + Y \leqslant 1)$.

Chapter 3.6 (P151)

Q2. Each student in a certain high school was classified according to her year in school (freshman, sophomore, junior, or senior) and according to the number of times that she had visited a certain museum (never, once, or more than once).The proportions of students in the various classifications are given in the following table:

	Never	Once	More than once
Freshmen	0.08	0.10	0.04
Sophomores	0.04	0.10	0.04
Juniors	0.04	0.20	0.09
Seniors	0.02	0.15	0.10

a. If a student selected at random from the high school is a junior, what is the probability that she has never visited the museum?

b. If a student selected at random from the high school has visited the

museum three times, what is the prob-ability that she is a senior?

Q3. Suppose that a point (X, Y) is chosen at random from the disk S defined as follows: $S = \{(x, y): (x-1)+(y+2)^2 \leqslant 9]\}$.

Determine (a) the conditional *p.d.f.* of Y for every given value of X.

(b) $\Pr(Y>0|X=2)$.

Q4. Suppose that the joint *p.d.f.* of two random variables X and Y is as follows:

$$f(x, y) = \begin{cases} c(x + y^2) & \text{for } 0 \leq x \leq 1 \text{ and } 0 \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Determine (a) the conditional p.d.f. of X for every given value of Y.

(b) $\Pr(X < 1/2|Y=1/2)$.